

Question 1:

Let the set of all values of r , for which the circles $(x+1)^2+(y+4)^2=r^2$ and $x^2+y^2-4x-2y-4=0$ intersect at two distinct points be the interval (α,β) . Then $\alpha\beta$ is equal to

- (A) 25
- (B) 21
- (C) 24
- (D) 20

Question 2:

Let the solution curve of the differential equation $xdy-ydx=\sqrt{x^2+y^2}dx, x>0$, with $y(1)=0$, be $y=y(x)$. Then $y(3)$ is equal to

- (A) 4
- (B) 2
- (C) 1
- (D) 6

Question 3:

Let the line $x=-1$ divide the area of the region $\{(x,y):1+x^2\leq y\leq 3-x\}$ in the ratio $m:n$, where $\gcd(m,n)=1$. Then $m+n$ is equal to

- (A) 27
- (B) 26
- (C) 25
- (D) 28

Question 4:

The number of solutions of $\tan^{-1}(4x)+\tan^{-1}(6x)=\pi/6$, where $-1/2\sqrt{6}<x<1/2\sqrt{6}$, is equal to

- (A) 1

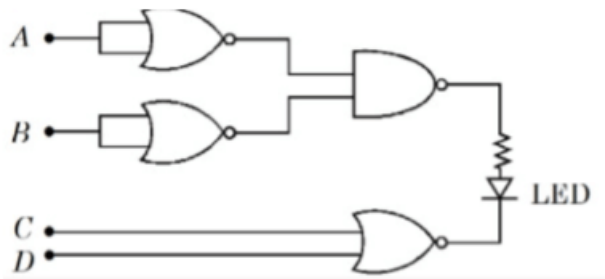
- (B) 2
- (C) 0
- (D) 3

Question 5:

Let ABC be a triangle. Consider four points p_1, p_2, p_3, p_4 on the side AB, five points p_5, p_6, p_7, p_8, p_9 on the side BC, and four points $p_{10}, p_{11}, p_{12}, p_{13}$ on the side AC. None of these points is a vertex of the triangle ABC. Then the total number of pentagons that can be formed by taking all the vertices from the points $p_1, p_2, \dots, p_{13}$ is _____.

Question 6:

Find the correct combination of A, B, C and D inputs which can cause the LED to glow.



- (A) 0100
- (B) 1000
- (C) 0011
- (D) 1101

Question 7:

Electric field in a region is given by $\vec{E} = Ax^i + By^j$, where $A = 10 \text{ V/m}^2$ and $B = 5 \text{ V/m}^2$. If the electric potential at a point (10,20) is 500 V, then the electric potential at origin is _____ V.

- (A) 1000
- (B) 500

- (C) 2000
- (D) 0

Question 8:

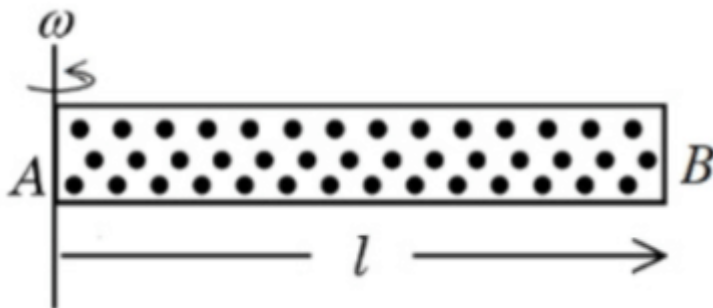
A simple pendulum has a bob with mass m and charge q . The pendulum string has negligible mass. When a uniform and horizontal electric field $\rightarrow E$ is applied, the tension in the string changes. The final tension in the string, when pendulum attains an equilibrium position is _____.

(g : acceleration due to gravity)

- (A) $\sqrt{m^2g^2 - q^2E^2}$
- (B) $\sqrt{m^2g^2 + q^2E^2}$
- (C) $mg + qE$
- (D) $mg - qE$

Question 9:

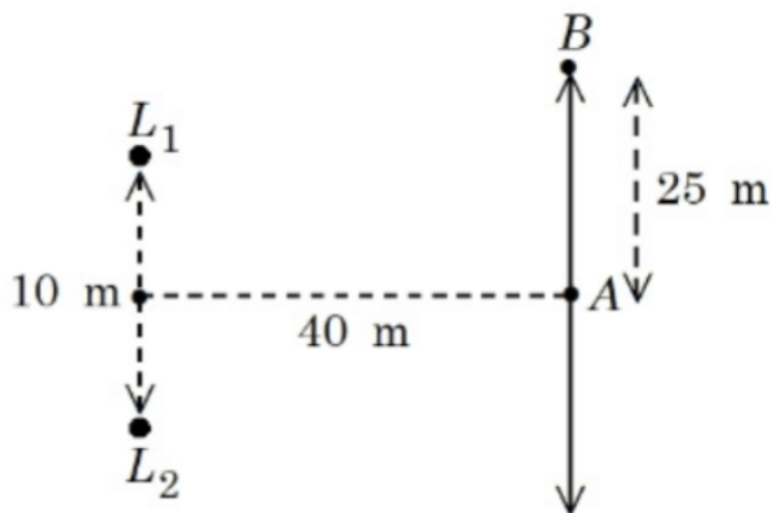
A cylindrical tube AB of length l , closed at both ends, contains an ideal gas of 1 mol having molecular weight M . The tube is rotated in a horizontal plane with constant angular velocity ω about an axis perpendicular to AB and passing through the edge at end A, as shown in the figure. If P_A and P_B are the pressures at A and B respectively, then (consider the temperature to be same at all points in the tube)



- (A) $P_B = P_A \exp(M\omega^2 l^2 / 2RT)$
- (B) $P_B = P_A$
- (C) $P_B = P_A \exp(M\omega^2 l^2 / 3RT)$
- (D) $P_B = P_A \exp(M\omega^2 l^2 / 2RT)$

Question 10:

Two loudspeakers (L_1 and L_2) are placed with a separation of 10m, as shown in the figure. Both speakers are fed with an audio input signal of the same frequency with constant volume. A voice recorder, initially at point A, at equidistance to both loudspeakers, is moved by 25m along the line AB while monitoring the audio signal. The measured signal was found to undergo 10 cycles of minima and maxima during the movement. The frequency of the input signal is _____ Hz.

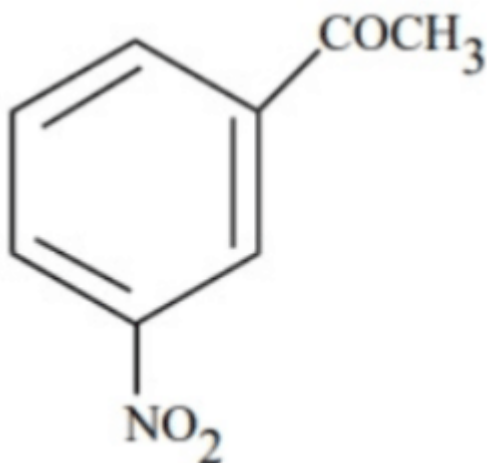


(Speed of sound in air is 324m/s and $\sqrt{5}=2.23$)

Question 11:

Given below are two statements:

Statement I: Benzene is nitrated to give nitrobenzene, which on further treatment with $\text{CH}_3\text{COCl}/\text{AlCl}_3$ will give the product shown.



Statement II: $-\text{NO}_2$ group is a meta-directing and deactivating group.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Statement I is correct but Statement II is incorrect
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is incorrect but Statement II is correct
- (D) Both Statement I and Statement II are correct

Question 12:

Two p-block elements X and Y form fluorides of the type EF_3 . The fluoride compound XF_3 is a Lewis acid and YF_3 is a Lewis base. The hybridizations of the central atoms of XF_3 and YF_3 respectively are

- (A) Both sp^2
- (B) Both sp^3
- (C) sp^2 and sp^3
- (D) sp^3 and sp^2

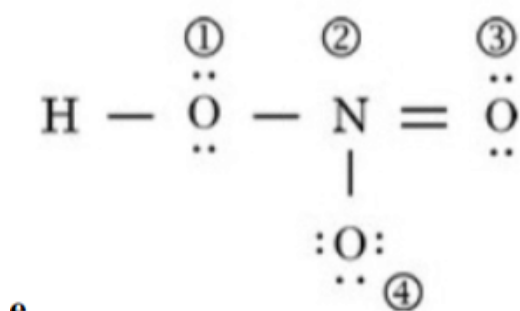
Question 13:

A p-block element E and hydrogen form a binary cation $(EH_x)^+$, while EH_3 on treatment with K_2HgI_4 in alkaline medium gives a precipitate of basic mercury(II) amido-iodide. Given below are first ionisation enthalpy values (kJ mol^{-1}) for the first elements each from groups 13, 14, 15 and 16. Identify the correct first ionisation enthalpy value for element E.

- (A) 1402
- (B) 801
- (C) 1312
- (D) 1086

Question 14:

The formal charges on the atoms marked as (1) to (4) in the Lewis representation of HNO_3 molecule respectively are



- (A) +1,0,0,-1
- (B) 0,-1,0,+1
- (C) 0,+1,0,-1
- (D) 0,0,-1,+1

Question 15:

Consider a solution of $\text{CO}_2(\text{g})$ dissolved in water in a closed container. Which one of the following plots correctly represents variation of $\log$ (partial pressure of CO_2 in vapour phase above water) [y-axis] with $\log$ (mole fraction of CO_2 in water) [x-axis] at 25°C ?